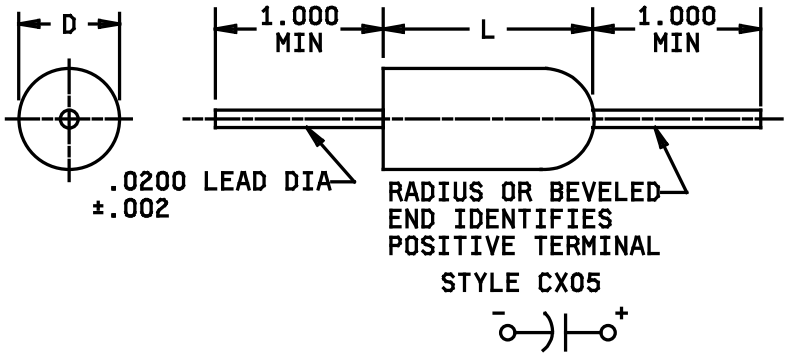
INCH-POUND
MIL-PRF-49137/5C
19 February 2009
SUPERSEDING
MIL-PRF-49137/5B
22 April 2004

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, MOLDED, NONHERMETICALLY SEALED, STYLE CX05

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49137.



CIRCUIT DIAGRAM

Case size	Dimensions	
	Diameter D ±.015 (0.38)	Length L ±.020 (0.51)
A	.085 (2.16)	.250 (6.35)
B	.100 (2.54)	.280 (7.11)
C	.170 (4.32)	.335 (8.51)
D	.170 (4.32)	.410 (10.41)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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TABLE I. Style CX05 characteristics.

Type designation 1/	DC rated voltage (85°C)	Capacitance (nominal)	DC leakage (max) +25°C	Dissipation factor	Case size
	Volts	µF	µA	%	
CX05D475-	6	4.7	1.0	6	A
CX05D156-	6	15	1.0	8	B
CX05D336-	6	33	1.5	8	C
CX05D476-	6	47	3.0	8	D
CX05F335-	10	3.3	1.0	6	A
CX05F106-	10	10	1.0	8	B
CX05F226-	10	22	1.5	8	C
CX05F276-	10	27	2.0	8	D
CX05F336-	10	33	3.0	8	D
CX05F396-	10	39	5.0	8	D
CX05F476-	10	47	5.0	8	D
CX05H225-	15	2.2	1.0	6	A
CX05H685-	15	6.8	1.0	6	B
CX05H156-	15	15	1.5	8	C
CX05H226-	15	22	3.0	8	D
CX05H336-	15	33	5.0	8	D
CX05J155-	20	1.5	1.0	6	A
CX05J475-	20	4.7	1.0	6	B
CX05J126-	20	12	1.0	8	D
CX05J156-	20	15	3.0	8	D
CX05K105-	25	1.0	1.0	4	A
CX05K225-	25	2.2	1.0	6	B
CX05K335-	25	3.3	1.0	6	B
CX05K685-	25	6.8	1.5	6	C
CX05K106-	25	10	1.5	8	C
CX05M334-	35	0.33	1.0	4	A
CX05M474-	35	0.47	1.0	4	A
CX05M155-	35	1.5	1.0	6	B
CX05M335-	35	3.3	1.5	6	C
CX05M395-	35	3.9	1.5	6	C
CX05M475-	35	4.7	1.5	6	C
CX05M685-	35	6.8	3.0	6	D
CX05M106-	35	10	5.0	8	D
CX05N104-	50	0.10	1.0	4	A
CX05N154-	50	0.15	1.0	4	A
CX05N224-	50	0.22	1.0	4	A
CX05N334-	50	0.33	1.0	4	B
CX05N474-	50	0.47	1.0	4	B
CX05N684-	50	0.68	1.0	4	B
CX05N105-	50	1.0	1.0	4	B
CX05N155-	50	1.5	1.5	6	C
CX05N225-	50	2.2	1.5	6	C
CX05N335-	50	3.3	2.0	6	D
CX05N475-	50	4.7	3.0	6	D

1/ Complete type designation will include an additional symbol indicating capacitance tolerance.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case material: Molded, epoxy or plastic.

Leads: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55°C to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Cap. tolerance: ±10% (K), or ±20% (M).

Dissipation factor (DF): See table I.

Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Within tolerance specified in table I.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Within -15 percent of step 1 measured value.
	DF	-	200 percent of table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	10 times 25°C limit (see table I).
	ΔCap.	-	Within +15 percent of step 1 measured value.
	DF	-	150 percent of table I.
Step 5 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.

Surge voltage: In accordance with MIL-PRF-49137.

DCL	-	See table I.
ΔCap.	-	Within 10 percent of initial value.
DF	-	See table I.

Life: Method 108 of MIL-STD-202.

1,000 hours:

At 25°C:	DCL	-	200 percent of table I.
	ΔCap.	-	Within 10 percent of initial measured value.
	DF	-	See table I.

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These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

NOTES:

Reference documents. In addition to MIL-PRF-49137, this specification sheet references the following documents.

MIL-STD-202

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

Changes from previous issues. The margins of this specification are marked with asterisks to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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